

3 **(long)** Figure 1 shows a diagram of a stationary turbojet engine with no bypass flow. Thermodynamic states are defined at five positions, labelled 1 to 5. It is to be modelled with the Joule cycle adapted for turbojet engines, assuming that the working fluid is a perfect gas, that the fuel mass flowrate can be neglected, that potential energy changes are negligible, and that the compressor, turbine, and exhaust nozzle are isentropic. The compression ratio is defined as $r_p = p_2/p_1$ and the non-dimensional heat addition is $\dot{Q}^* = \dot{Q}/(\dot{m}c_pT_1)$, where $\dot{m}$ is the air mass flowrate. It is helpful to define $\eta = (\gamma - 1)/\gamma$.

(a) Starting from an appropriate Tds equation, show that lines of constant p on a $T-s$ diagram for a perfect gas have the form $T/T_0 = \exp((s - s_0)/c_p)$. [2]

(b) Show that $T_4/T_1 = 1 + \dot{Q}^*$ for the stationary turbojet engine. [3]

(c) On the same $T-s$ diagram, and with the same values for T_1 and $\dot{Q}^*$, sketch the thermodynamic cycles of the stationary turbojet for

- (i) a high compression ratio (label T_1, T_2, T_3, T_4, T_5);
- (ii) a low compression ratio (label $T'_1, T'_2, T'_3, T'_4, T'_5$).

Pay particular attention to the sizes of $(T_2 - T_1)$, $(T_3 - T_2)$, and $(T_4 - T_3)$ at both compression ratios, as well as to the locations of T_4 and T'_4 . [8]

(d) The jet Mach number is defined as $M = V_5/\sqrt{\gamma RT_5}$. Show that $(\gamma - 1)M^2 = 2((p_4/p_5)^\eta - 1)$. Sketch M as a function of p_4/p_5 . [8]

(e) Derive an expression for p_4/p_1 as a function of r_p and $\dot{Q}^*$. [6]

(f) Using the $T-s$ diagrams in part (c), explain why the jet velocity increases with r_p for a given $\dot{Q}^*$. [3]

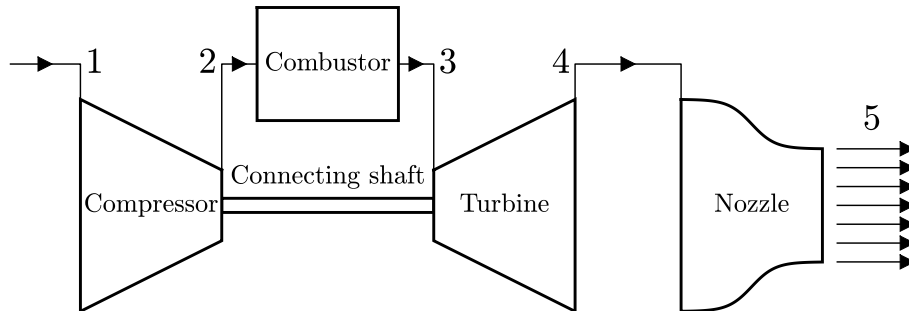


Fig. 1